Safe System of Work

Project:-

Location:-

Date:-

TRE ENERGY LTD
Unit 2, Aghanloo Industrial Estate
Limavady
BT49 0HE

Project/Contract	
Client	
Location	
Start Date	
Expected Duration Date	

Emergency Contact Details			
Contact **Tel** **Mobile**	John Adams TRE Energy Ltd 00447377444122		Fire/Emergency/Police 999

Supervisor on Site	**Name**	**Contact Telephone Number**

Approximate Number of Operatives on Site Each Day	

Data Protection Statement
The information and data provided herein shall not be duplicated, disclosed or disseminated by the recipient in whole or in part for any purpose whatsoever without the prior written permission from TRE Energy Ltd.

Disclaimer
In carrying out this assignment, we have taken all reasonable steps to ensure that those hazards that could be reasonably expected to cause harm have been identified. However, it is important to remember that ongoing assessments will be required as new situations and conditions are encountered and new processes introduced.

This documentation is based upon analysis of information provided by the representatives of the owners of the facility. The document author therefore makes no warranty expressed or implied as to the accuracy of this data or opinions.

Operatives Information Sheet

The following worksheets have been developed to provide a safe system of work and must be adhered to at all times, any significant deviation from this system must first be authorised by your Director, John Adams & the Client.

Please read the entire sheet before beginning the task, if you have any questions please contact your manager or safety representative.

The main hazards to your safety and health are

a) Falling from Height
b) Injury from slips trips and falls
c) Injury from Manual handling
d) Access/Egress
e) Injury due to use of handheld tools/equipment
f) Exposure to hazardous substances
g) Electricity
h) Noise
i) Dust
j) Exposure to Inclement Weather
k) Fire
l) Vibration
m) Crane Operations
n) Teleporter operations
o) Working on a public building/presence of members of the public
p) Roof work
q) Scaffolding/Edge Protection
r) Confined space work

The main hazards to the environment are

a) Waste produced from the task
b) Dust
c) Noise

Preventative Measures you must take

a) You must be “competent” to carry out the task.
b) You must NOT work alone.
c) You must ensure the area is safe to work in and left safe at the end of each working day.
d) You must ensure the safety of others at all times who could be affected by your work operations.
e) You must not lift beyond your capabilities.
f) You must follow all directions given by the Company, Client, and appointed Supervisor.

Personal Protective Equipment you must wear

a) Hard Hats
b) Hi Visibility Vests
c) Safety boots
d) Eye protection (when using power tools such as drills)
e) Gloves (when manual handling and using hazardous substances)
f) Safety Harness & Lanyard to anchor if required
g) Ear protection (when using power tools)
h) Dust masks (FFP3 masks to be worn when cutting)

Environmental Protection Measures you must take

All waste must be segregated and removed to the allocated waste skips or your company vehicle and removed off site for disposal.
Dust emissions must be controlled. All cutting tools must have extraction at source and materials cut externally in the open air so far as it is reasonably practicable to do so.

Noise levels must be kept to a minimum so far as it is reasonably practicable and monitored.

Quality Control

a) Adhere strictly to the following procedure to ensure quality of service
b) If in doubt contact the Company for clarification before proceeding.

General Precautions

Operatives, Training & PPE
This project will be carried out by operatives of TRE Energy Ltd and competent appointed sub contractors. All operatives and persons appointed have completed their 1 Day CSR course/CSCS Training Course, work at height awareness training course and other relevant training courses.

P.P.E worn must include, safety footwear, safety vest, safety glasses, safety hard hat, safety masks (FFP3), ear protection and safety harness and lanyard when working on a roof (if this is required).

Preparation & Induction
Operatives & appointed sub-contractors will be inducted onto this project by their Employer (TRE Energy Ltd) & the Client and will follow all site rules and safety procedures.

All appointed sub-contractors must ensure that they have briefed their operatives prior to start of work.

Communication of Information
All appointed operatives and sub-contractors will be inducted prior to start of works. Daily briefing will take place and regular updates will be communicated as and when required.

Welfare
Operatives will be permitted to use the temporary toilets made available. All operatives must have drinking water (bottled) with them and ensure they stay hydrated at all times. Hands must be washed before consuming food or beverages.

First Aid
A member of the team will have basic emergency first aid training. A fully stocked first aid kit will be retained in the company vehicle. In the event that a persons is injured or gets into difficulty, the appropriate authority will be contacted via a mobile phone with immediate effect and all instructions given by the appropriate authority followed.

Work is being carried out on a live hospital site so in the event of an emergency/accident that requires medical attention, operatives must go direct to A&E.

Access egress
Safe access and egress to the work area will be maintained at all times. Access to the roof will be via scaffold that will be erected, inspected and maintained throughout the duration of the project by a competent appointed sub-contractor.

Material Handling
All materials required will be brought to the work area manually and transported to the roof via a crane (this will be operated by a competent contractor) or a teleporter (operated by a competent operator). Materials will be loaded on to a loading bay and manually moved on to roof.

Manual Handling
All operatives have been instructed on the potential dangers of manual handling and operatives will not lift items of equipment beyond their capabilities. Heavy or awkward items will be broken down into smaller pieces or dual lifted where this is not possible. It is the responsibility of the Company, Client & site supervisor to identify and control potentially dangerous manual handling situations as they occur on a day to day basis.

The following is provided for general guidance only when required to carry out a manual handling task:

1. Size up the job, remove obstructions, make sure there is a clear space where the load has to be set down. Ensure you can see over the load when carrying it.

2. Stand close to the object and with feet up to 8 to 12 inches apart, one foot in advance of the other, prepare to lift.

3. Bend your knees into crouch position, back straight (not necessarily vertical).

4. Pull your chin in and avoid dropping your head forward.

5. A good grip is required, preferably one hand around the front of the load and one hand underneath to prevent the load slipping forward or down.

6. Pull the object close to the chest, it is easier to handle and there is less chance of it slipping.

7. An even lift is required to complete the exercise.

Working at height
Falls from height remain the single biggest cause of death in the construction industry.

All operatives are aware of the dangers and consequences of falling from height, all working at height will be supervised.

Scaffold will be erected by a competent appointed subcontractor, and it will be their duty to inspected it weekly, sign off scaf tags and make any changes as required.

Access to the roof will be via scaffold and scaffold will be erected in accordance with the requirements and regulations and industry standards.

If weather conditions become adverse (unsafe to work in), all work will be stopped until weather conditions improve and it is safe to recommence.

Working on a Roof
Operatives trained in work at height and working on a roof. Access will be via a scaffold which will be fitted with edge protection and harnesses and lanyards will be attached and worn where it is required.

Roof has been structurally assessed and approved to access safely.

Always ensure that you have a clear view of all areas of the roof and there are no openings covered over etc with moss/greenery for example.

Vehicle Control/Parking
All site operatives are required to park in a safe manner in the designated areas marked out in the car park. This will be pre agreed with the Client and an area set aside for vehicles.

All operatives are expected to adhere to the highway code requirements and ensure that they operate vehicles in a safe manner and always ensure the safety of the public and other drivers.

Drivers to obey the traffic signs and traffic systems in place throughout the hospital grounds.

Noise
Noisy activities will be carried out between the agreed hours of 9am -5pm ,Monday to Friday. All operatives will wear their hearing protection when using noisy tools/equipment.

Fire
The risk of a fire starting is very low. Operatives will have fire extinguishers (fully charged and tested) in their company vehicle. Operatives are competent in what to do in the event of a fire.

Smoking is not permitted; all electrical portable equipment is annually P.A.T tested. The use of electronic smoking materials is not permitted.

In the event of a fire, operatives must evacuate to a safe area and call the appropriate authorities.

Occupational Health
All operatives are provided with the appropriate information on occupational health risks associated with their scope of works and work industry.

Dust will be minimised so far as is reasonably practicable and when cutting or drilling, FFP3 masks will be worn by all operatives.

All operatives are provided with the appropriate ear protection which they must wear when using noisy equipment.

Vibration tools being used must not be used for any longer than is deemed safe.

The Company promote a positive mental health in the workplace and communicate this via signage, tool box talks and newsletters.

Working on a Public Building
Care will be taken to ensure that the public are protected at all times. The area below where work at height is being carried out and the area around the base of the scaffold will be protected and there will be walkways and signage provided.

Signage must be in place to advise all members of the public that works is being carried out.

Crane Operations
A crane will be provided and operated by a competent contractor and it will be inspected as per statutory requirements.

The base around the crane will be made secure.

Loads will not be lifted above people and a banksman will be present to assist the crane operator.

*Note:- The crane contractor and scaffold contractor will be responsible for conducting their own risk assessments, writing their own methods of work and devising a lift plan prior to start of works.

Teleporter Operations

In the event that a teleporter is used to lift panels onto scaffold/roof, it must be operated by a trained and competent operator only.

The SWL must not be exceeded, and all teleporters must be checked before use and maintained as per statutory requirements.

It is also required that a banksman assists the operator and loads are not lifted above or close to public access/walking/parking areas.

Accessing a Confined Space

When accessing a confined space to install and connect panels to distribution boards, care must be taken to ensure that safe access and egress is available or provided and maintained.

Scaffold

Scaffold will be erected by a competent contractor however you must ensure that it has been inspected before you access it and the scaf tag is completed and attached.

DO NOT MAKE ANY ADJUSTMENTS to scaffold.

Ensure that the area around it is isolated with fencing or similar and all ladders tied off/covered at the end of each day.

Method Statement: Solar PV Installation on Trapezoidal Metal Roof

Project Description

Main Elements of the Work

1. Arrival and Site Preparation

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2. Site Coordination and Briefing
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3. Work Area Setup
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4. Pre-Start Equipment and Safety Checks
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5. Skylight Mesh Installation (Fall Protection)

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6. Material Handling

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7. Mounting System Installation

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8. Solar Panel Installation

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9. Electrical Works

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10. Testing and Commissioning

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11. Completion and Site Clearance

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Additional Notes

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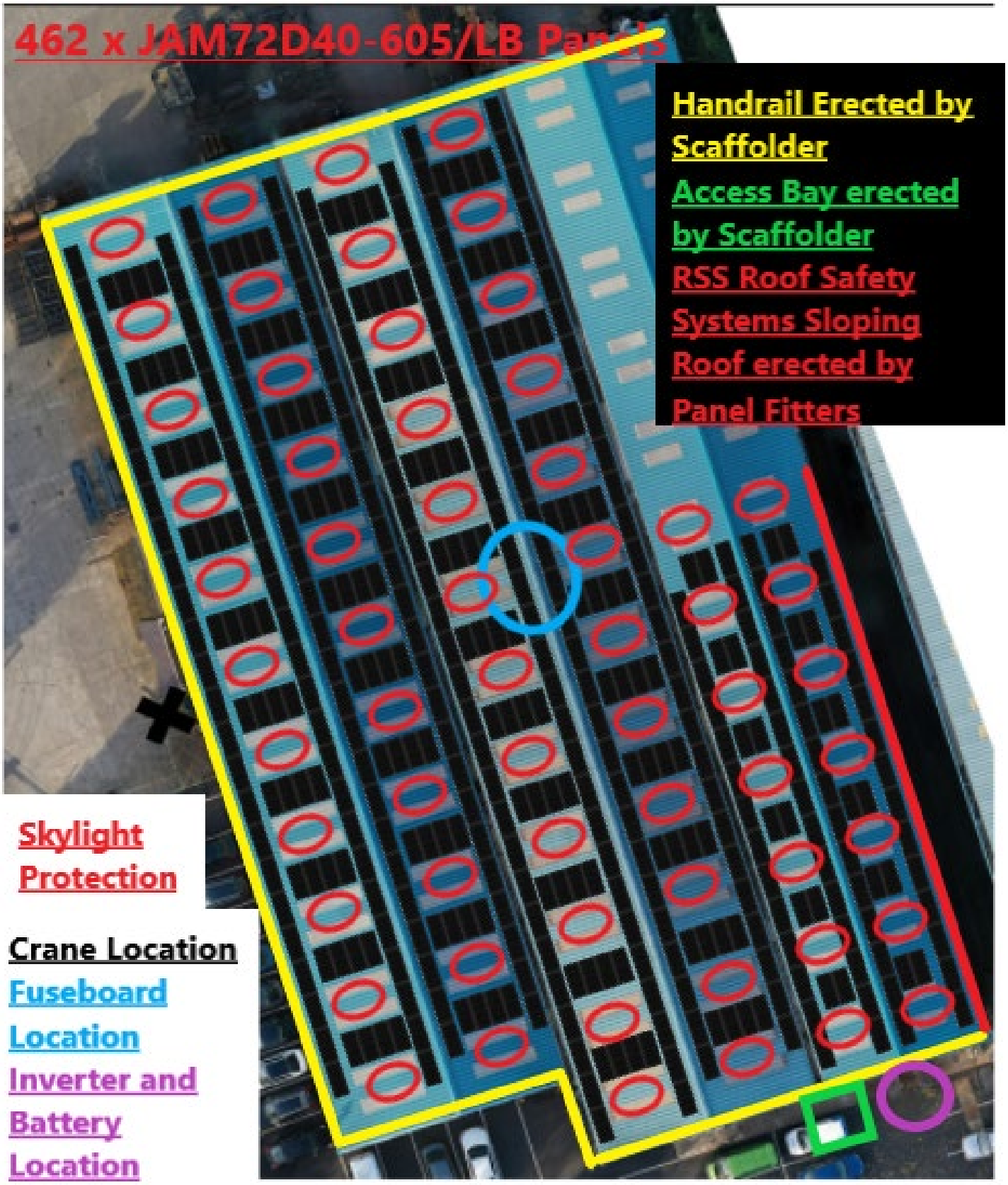
LARGE SHED


462 x JAM72D40-605/LB Panels
Handrail Erected by Scaffolder
Access Bay erected by Scaffolder
RSS Roof Safety Systems Sloping Roof erected by Panel Fitters
Skylight Protection
Crane Location
Fuseboard Location
Inverter and Battery Location

SMALL SHED

132 x JAM72D40-605/LB Panels
Fuseboard and Inverter Location
Hand rail erected by Scaffolder
Access Bay erected by Scaffolder
Skylight Protection

Material Delivery List

Line No.	Category	Description	Job Total QTY	Delivery QTY (If different)	Checked?
1	Panel	Trina Solar (600 Watt) - 2382x1134mm	594		[ ]
2	Panel	Crane	1		[ ]
3	Panel	Concrete base and ducting	1		[ ]
4	Hybrid Inverters	Solis Hybrid 50k and 1Nr 80kw solis 3 phase string inverter	4 and 1		[ ]
5	Batteries	(NL) ECO-E107WS = Air-cooled Hybrid Solar ESS Cabinet	3		[ ]
6	Roof	End clips	362		[ ]
7	Roof	Mid Clips	1154		[ ]
8	Roof	Panel Cable (per meter)	5940		[ ]
9	Roof	2 Tack	1485		[ ]
10	Panel Electrics	Three phase generation meter	1		[ ]
11	Panel Electrics	G99 Protection Relay	1		[ ]
12	Panel Electrics	Solis Wifi Stick	1		[ ]
13	Scaffolding	2 Storeys 60m Lng	1		[ ]
14	Civils/Ducting	20m Duct Through Concrete - To Panels	1		[ ]

Risk Assessment

Activity	Risk	Action
Manual Handling	Musculoskeletal Disorders Sprains/Strains Back Injuries Cuts/Abrasions	All operatives are aware of the risk of incorrect manual handling. Manual handling will be kept to a minimum. Operatives will be advised not to hold or maintain awkward positions. No overreaching or over stretching. Panels etc and heavier materials/equipment will be brought to the work area via a crane. All persons must wear suitable footwear and gloves. Weights of materials must be made known so far as is reasonably practicable.
Use of hand held Equipment	Cuts/Abrasions Superficial Injuries Splashes	Equipment to be checked before use to ensure safe to use. Safety gloves, eye protection will be worn at all times. All operatives competent within their role. No over reaching or over stretching when using equipment in work platform. Any defectives to equipment to be reported immediately. Equipment cleared away after use. Spillages must be cleaned up immediately.
Slips and Trips	Personal injury Injury to Others	High standards of housekeeping will be maintained at all times. Walk ways will be kept and maintained in a safe manner at all times. Work equipment and materials etc will be kept in secure area/vehicle until required for use. Leads must not be trailed across walkways. All site operatives must ensure the work area is kept and maintained in a safe manner at all times.

Transporting materials/equipment to work area.	Slips, Trips & Falls Public Injury	Safety Footwear will be worn, gloves will be worn. A teleporter will be used to move materials required for this project. The area below where work is being carried out will be isolated and maintained in a secure manner at all times. Banksman will direct movement of materials on the ground. This project will be supervised.
Use of handheld tools/Cordless drills/Saws	Cuts/Abrasions Superficial Injuries	All tools to be checked before use to ensure safe to use. Cutting will be carried out away from the body. Safety gloves, ear protection, eye protection and dust masks will be worn at all times. Faulty equipment will not be used- will be taken out of use immediately and replaced. Guards will be in place and not adjusted in anyway. All cutting will take place on a solid/supported surface.
Working on a Roof	Collapse Falls Access/egress Covered openings Structural safety Injury to others Adverse weather conditions	A competent structural assessor will conduct a survey to ensure that the roof is safe to access. All openings will be highlighted and made safe. Access and egress will be via scaffold. If weather conditions are adverse, work will cease until it is safe to recommence.
Scaffold	Collapse Falls Falling materials Access/egress Lack of inspection protocol Removing parts Injury to the public Unauthorised access Overloading	Scaffold erected, inspected and maintained by competent, appointed scaffolder. Operatives aware that they must not remove any parts or make any adjustments. Access and egress via fixed and tied ladders. Ladders secured at the end of each working day to prevent unauthorised access. Edge protection in place. SWL not exceeded. Ground conditions suitable.

		All operatives trained in work at height. Walkways covered over and signage in place to direct public and alert the public of works. Loading bays will be double boarded.
Working on a live building	Injury to others Damage to the building Interrupting day to day activities/operations	Daily contact with the Client will be maintained to ensure that no issues arise. Signage in place to advise the public of works being carried out. Scaffold, crane and all work areas will be maintained in a safe and secure manner. All public walkways and roadways will be kept clear at all times. Any problems that arise will be dealt with with immediate effect.
Working outside	Adverse weather conditions Presence of overhead services Poor visibility Hypothermia	Work will be carried out during day light hours when visibility is good. The presence of overhead services is not anticipated. Work will cease if weather conditions become adverse. Operatives to wear thermal clothing underneath all P.P.E. Operatives to take regular breaks indoors and consume hot drinks.
Electricity	Shock Fatal injury Fire Damage to building Injury to others	All electrical work will be carried out by competent electricians. Working on live systems not permitted. Battery operated tools will be used. Chargers P.A.T tested as per statutory requirements. Fire extinguishers available in all company vehicles.
Presence of Other Contractors / Members of the Public	Injury to members of the public / Other Contractors	Work will be supervised at all times. Area below and around where work is being carried out will be isolated at all times, signage in place and barriers in place. Plant operated by trained and competent people only. Scaffold erected will have covered and protected walkways with signage in place to direct the public.

Dust	Respiratory irritation and related diseases	All cutting will take place outside. Operatives will have equipment with extraction at source and water devices to dampen down materials as required. FFP3 masks are provided and must be worn at all times when cutting or drilling.
Fire	Smoke Flames Heat Personal injury/fatalities Structural damage	All portable electrical tools will be P.A.T tested prior to being used on site. Fuelling of plant will be conducted off site. Smoking prohibited. Extinguishers provided. All site operatives made aware of fire safety via induction.
Working at Height	Falls Falling materials Unsafe access and egress	Safe access and egress provided and maintained to areas off the ground (scaffold e.g). Areas around where work at height is being carried out will be isolated and cordoned off. All roofs will be structurally approved to work on before access permitted. Ladders used for access will be of industrial strength and tied and footed appropriately. All persons working at height will be competent. Harness and lanyard will be used as required. Edge protection provided where required. Weather conditions must be suitable for work at height to be carried out safely. P.P.E worn including hard hat with chin strap.
Access and egress	Unsafe access and egress Trips, slips, falls Areas hard to access/egress	Suitable and safe access and egress provided and maintained. Good standards of housekeeping maintained to avoid slips, trips, falls. Access and egress will be monitored to ensure unauthorised access not permitted. Safe means of access and egress provided and maintained.

Noise	Hearing problems. Nuisance for others. Inability to hear instructions/warnings.	Noisy activities carried out between the hours of 08.00am-17.00hrs. Ear protection worn when using tools/equipment or operating plant whereby the noise levels exceed 85 dB(A). If any person is experiencing hearing problems they must contact TRE Ltd.
Inclement weather	Adverse weather conditions Hypothermia Sunstroke Sun burn Vision impaired Damage to materials Material movement Icy/snowy conditions – slips, trips, falls	Work will only be carried out when weather conditions are suitable to work in safely. During colder months all operatives must wear thermal clothing, take regular breaks and consume hot fluids regularly. Access to height must not be conducted if there is high winds, icy conditions, snow etc. Sun protection to be worn when working outside. During summer months, regular cold drinks must be consumed.
Operation of Teleporter	Untrained operator Lack of maintenance Lack of Inspection SWL exceeded Falling loads Collision with others/vehicles/buildings Refuelling Accessing moving parts Reversing/Lifting Loads	Teleporter will only be operated by competent persons who have been trained. Maintained as per statutory requirements. Switched off and keys removed when not in use. Defects must not be fixed by anyone other than a competent plant repair mechanic. Inspected visually only weekly when power isolated. SWL adhered to. Loads secure. No lifting above where others are walking. Banksman available to direct driver. Warning devices operating correctly and used to alert others. No operated in public areas/area being operated isolated. Refuelled safely and spillages cleaned up as required. No smoking or using mobile phone when refuelling.
Confined Space Access	Unsafe access Lack of	Confined spaces will be ascertained first before commencement of job to

	light/ventilation/movement Poor housekeeping Falling Lack of unsafe walkways	ensure that they are accessible. If required suitable industrial strength ladders will be used to access height. Artificial lighting provided if required. Walkways in the confined spaced must be available. Any areas out of bounds due to risk of falling through must be made known by Client and out of bounds.

Declaration

I have read and had explained to me the contents of this safe system of work and will not deviate away from it under any circumstances.

I understand the contents and will work in accordance with this safe system of work at all times.

I will not do anything that could compromise my safety or the safety of others.

MAKE SURE THAT YOU SIGN OFF ON THE DIGITAL SAFETY SYSTEM

PRINT NAME	SIGNATURE	DATE